

1 LLama2 运行结果

```
(base) jzd@DESKTOP-75609C8:~/lab6/llama2.c$ ./run stories15M.bin
Once upon a time, there was a kind dog named Spot. Spot loved to play with his friends in the park. One day, Spot saw a little girl with a pretty bow . He went up to her and said, "Can I play with your bow?"
The little girl smiled and said, "Sure, you can play with my bow." Spot was so happy and wagged his tail. But then he accidentally dropped the bow and it rolled away. The little girl was sad because she loved that bow.
Spot felt bad for dropping the bow and said, "I'm sorry, I didn't mean to drop it." The little girl was happy that Spot said sorry. She knew he was always polite and kind. They continued to play together and have fun in the park. The end.
achieved tok/s: 129.533679
(base) jzd@DESKTOP-75609C8:~/lab6/llama2.c$ ./run stories42M.bin
Once upon a time, there was a little boy named Tim. Tim had a brother named Sam. They lived in a small house with their mom and dad. One day, it was very dark outside. Tim and Sam wanted to play, but it was too dark to see.
Tim had an idea. He said, "Sam, let's make our own fun in the dark!" Sam agreed, and they got some chairs and blankets. They used the chairs and blankets to make a big tent in their room. They played inside the tent, pretending it was their town.
As they played, Tim said to Sam, "I can't see far, but we can still have fun!" Sam smiled and replied, "You're right, Tim. We can still have fun even if it's dark outside." They played all night in their little tent, laughing and being together.
achieved tok/s: 42.395693
(base) jzd@DESKTOP-75609C8:~/lab6/llama2.c$ ./run stories110M.bin
One day, a little boy named Tim went to the park with his mom. They saw a big tree with many leaves. Tim wanted to pick a leaf for his mom, but he was too small. He felt sad.
Then, a tall man came to the park. He saw Tim looking at the tree. The man asked, "Do you need help, little boy?" Tim said, "Yes, I want to pick a leaf for my mom, but I can't reach."
The tall man smiled and picked a leaf for Tim. Tim was very happy. But when the man gave the leaf to Tim, Tim saw that it was fake. It was not a real leaf. Tim was surprised and said, "This is not a real leaf!"
The tall man laughed and said, "I'm sorry, little boy. Let's go find a real leaf for your mom." Tim and the tall man looked around the park and found a real leaf on a lower tree. Tim's mom was very happy when she saw the leaf. They all laughed and had a great day at the park.
achieved tok/s: 15.729562
```

模型越大，运行速度越慢。

2 LLama2 优化

采用 AVX 进行优化，

优化过程中使用了 AVX 指令对矩阵乘法过程中的内积部分进行了向量化处理。

通过使用 AVX 提供的 mm256_loadu_ps 和 mm256_add_ps 等指令，将每次计算从单一标量扩展为 8 个浮点数的并行操作。利用 AVX 指令加载、计算、存储数据，每次处理 8 个浮点数；在最后的的部分，针对剩余元素进行了逐元素计算。

```
void matmul(float* xout, float* x, float* w, int n, int d) {
    int vector_width = 8;

    #pragma omp parallel for
    for (int i = 0; i < d; i++) {
        float val = 0.0f;

        int j;
        __m256 sum_vec = _mm256_setzero_ps();

        for (j = 0; j + vector_width <= n; j += vector_width) {
            __m256 w_vec = _mm256_loadu_ps(&w[i * n + j]);
            __m256 x_vec = _mm256_loadu_ps(&x[j]);
            sum_vec = _mm256_add_ps(sum_vec, _mm256_mul_ps(w_vec, x_vec));
        }

        float sum_array[8];
        _mm256_storeu_ps(sum_array, sum_vec);
        val = sum_array[0] + sum_array[1] + sum_array[2] + sum_array[3] +
            sum_array[4] + sum_array[5] + sum_array[6] + sum_array[7];

        for (; j < n; j++) {
            val += w[i * n + j] * x[j];
        }

        xout[i] = val;
    }
}
```

优化之后，进行测试：

```
(base) jzd@DESKTOP-75609C8:~/lab6/llama2.c$ ./run stories15M.bin
Once upon a time, there was a little boy named Timmy. Timmy loved to play with his toy jeep. One day, Timmy and his mom went to the park. Timmy brought his jeep with him.
While they were walking, Timmy accidentally dropped his jeep. "Oh no!" cried Timmy. "My jeep is broken!"
His mom asked, "Don't worry, Timmy. We can buy you a new one. But this one is very expensive."
Timmy was happy and said, "Yay! Thank you, Mommy!"
They went to the store and bought a new jeep. Timmy was so excited to play with it. He drove it all around the park, making vroom vroom sounds. Timmy had so much fun with his new jeep.
achieved tok/s: 242.837653
(base) jzd@DESKTOP-75609C8:~/lab6/llama2.c$ ./run stories42M.bin
Once upon a time, there was a little girl named Lily. She loved to play outside and explore the world around her. One day, she went to the pond to look at the fish swimming. She saw a big, dirty fish swimming in the pond.
Lily said, "Hello, fish! Why are you so dirty?"
The fish replied, "I like to swim in the dirty water."
Lily said, "But your tail is dirty. You should go to the spray pot."
The fish said, "What is a spray pot?"
Lily said, "It is where you can spray water on your tail. Then it will be clean."
The fish thought about it and said, "Thank you, Lily. I will go to the spray pot." And so the fish swam away to the spray pot, happy to be clean again.
achieved tok/s: 93.949828
(base) jzd@DESKTOP-75609C8:~/lab6/llama2.c$ ./run stories110M.bin
Once upon a time, there was a girl named Lily. She had a big red balloon that she loved very much. One day, she went outside to play with her balloon. She was having so much fun until she accidentally let go of it and it flew away. Lily was very sad because she loved her balloon.
Her mom saw how sad she was and wanted to make her happy again. She told Lily to close her eyes and hold out her hand. Then, her mom put a special string on her palm. When Lily opened her eyes, she saw that her hand was full of balloons! Her mom had supplied her with lots of balloons to make her happy again.
Lily was so happy that she jumped up and down, but she was also a little embarrassed because she had let go of her balloon. She promised herself to be more careful with her balloons in the future.
achieved tok/s: 35.821422
```

可见，相比之前的运行速度有了很大的提升。

多次测试，得到

	1	2	3	avg
stories15M.bin	242.84	241.99	242.51	242.45
stories42M.bin	93.95	93.88	93.90	93.91
stories110M.bin	35.82	35.88	35.67	35.79

相较于优化之前，stories15M.bin 速度提升为原来的 187.6 %，stories42M.bin 提升为原来的 221.5%，stories110M.bin 提升为原来的 227.7%。